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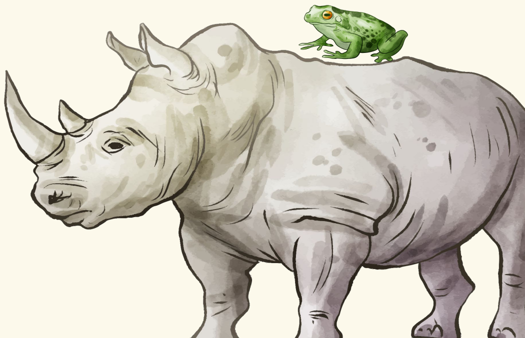
EXTINCT ANIMAL TRACKER

This project aims to create a tool that visualizes data on extinct and endangered species to raise awareness and support conservation efforts. By analyzing historical and current data, we hope to highlight patterns in species extinction and help predict future trends



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01. Background and Motivation

"Be kind to animals, great and small, for they deserve our care, one and all". Animals, the most significant aspect of our life, are disappearing from the planet we live in today. Is it our fault? How are they being put in danger? How can we help them to survive? are a few of the queries that our visualization project, "Endangered Species," would try to address. The purpose of this initiative is to raise public awareness of animal welfare issues and the causes underlying them. Many species are on the verge of extinction due to habitat destruction, climate change and pollution. Loss of animals eventually hamper the food chain and affects the entire system negatively. According to the IUCN Red List, more than 41,000 species are now at risk of extinction, with this number increasing daily.

02. Project Objectives

The primary goal of this project is to create an interactive, web-based tool that visually tracks and highlights the conservation status of endangered species worldwide. This tool will offer users the ability to explore species data in an intuitive, engaging manner, providing insights into the factors driving species endangerment and the global patterns of biodiversity loss. By utilizing data from credible sources such as the IUCN Red List, the project aims to provide a platform where users can explore the current state of species under threat, understand the causes behind their endangerment, and track changes in their conservation status over time.

03. Rough Ideas created:

1.

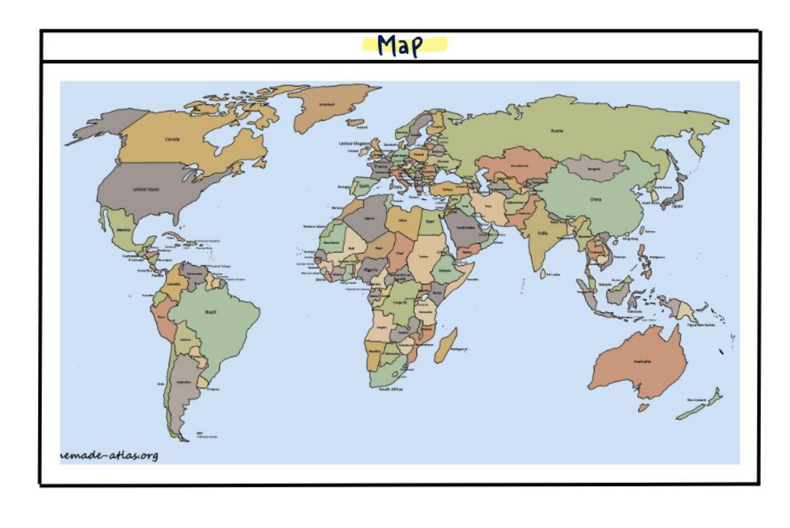


Fig.1

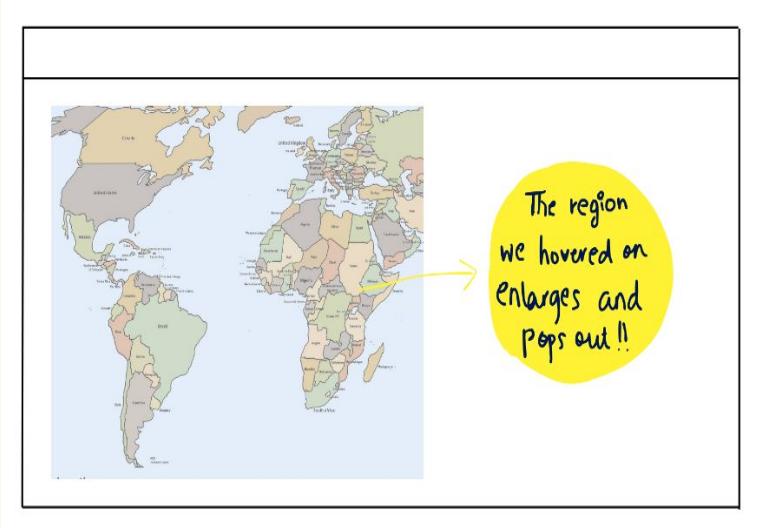
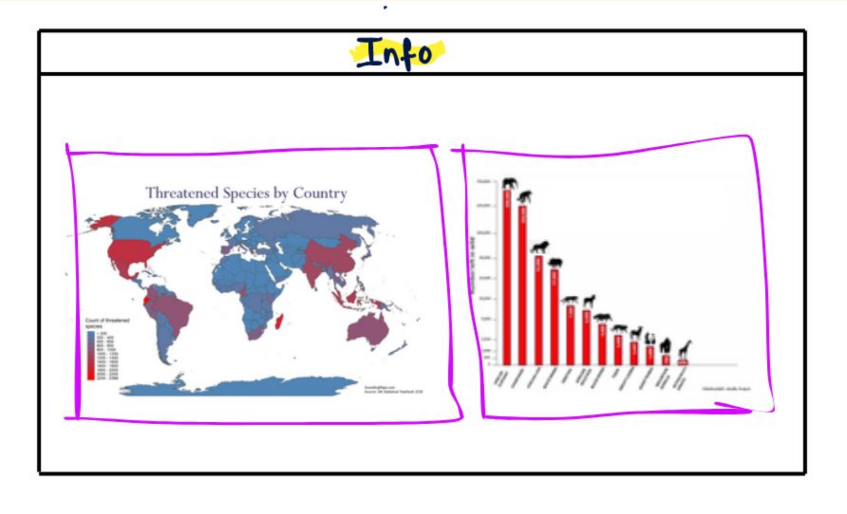


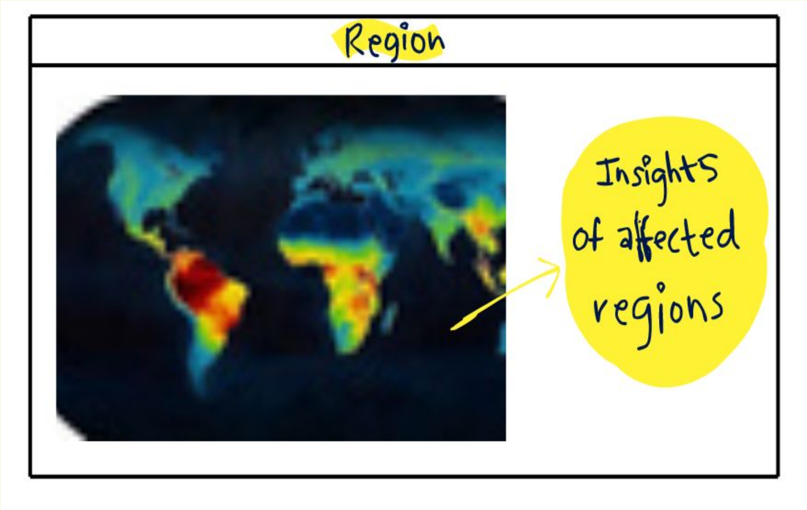
Fig.2

2.



Info
Threatened Species by Country

Fig.3



Region
Insights of affected regions

Fig.4

04. Data Source Identification

Our project began with an in-depth analysis of potential data sources for tracking extinct and threatened species. After thorough research, we identified two primary resources:

1. IUCN (International Union for Conservation of Nature), which maintains the widely recognized "Red List" of threatened species.

2. GBIF (Global Biodiversity Information Facility), another valuable data repository, which we plan to explore for future visualizations.

05. Accessing Data via IUCN API V

Once we finalized IUCN as our data source, we explored the IUCN API V4. This API provides access to a vast array of taxonomic data and assessments of species, which inspired us to create visualizations focusing on global extinction patterns. One visualization idea was a heatmap displaying the countries with the highest number of species facing extinction.

06. Initial Development: React and D3

To bring our idea to life, we started by building a small React app. After setting up our environment with Node.js and React.js, we began rendering the map using D3.js. Though we initially made good progress, D3's complexity proved challenging, especially for our intended visualizations. After consulting a TA, we decided to transition to Chart.js, which provided a simpler yet effective way to handle charts and maps.

07. Data Processing Challenges

During the data exploration phase, we faced a few hurdles:

1. The IUCN API did not offer a single endpoint that provided a direct list of threatened species by country. Instead, it returned multiple assessments requiring extensive processing.

2. We shifted focus from the API and started exploring spatial datasets that matched our map-based visualization needs. This led us to .shp (Shapefile) formats, which contained the geospatial data we required.

08. Data Conversion Using QGIS

To integrate the spatial data into our visualizations, we used QGIS (a free GIS software) to convert the shapefiles into GeoJSON format, which is compatible with mapping libraries. However, when feeding the GeoJSON data into Chart.js, we encountered numerous errors. These issues made us reconsider our data approach.

09. Obtaining Country-Level Data from IUCN

In search of a more efficient method, we directly contacted IUCN to request the data. They provided us with a CSV file containing country-wise data for threatened species. This file became the foundation for our current map visualization.

10. Data Transformation and Mapping

To process the country-wise data:

1. We converted the CSV file into JSON format, making it easier to manipulate within our app.

2. Chart.js required ISO 3166 country codes for accurate mapping of the species data. However, our dataset lacked these codes. To resolve this, we downloaded a separate JSON file of country codes and matched it to our species data.

3. Using a hash map, we linked the country codes to the respective number of threatened species.

11. Visualization: Chart.js with Red Hue Scale

With the processed data in place, we finalized the visualization using Chart.js. We applied a red hue scale to highlight countries with the highest extinction rates. The color intensity reflects the severity of the threat, helping users quickly identify regions most impacted by species extinction.

12. Future Enhancements

Although we successfully created the current visualization using Chart.js, we plan to revisit D3.js in the future for more sophisticated and flexible visualizations. The current project serves as a solid foundation for ongoing improvements and new insights into the global extinction crisis.

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